TECHNO INDIA SILIGURI INSTITUTE OF TECHNOLOGY
A Satyam Roychowdhury Initiative

Looping Statement

For, while and loop control

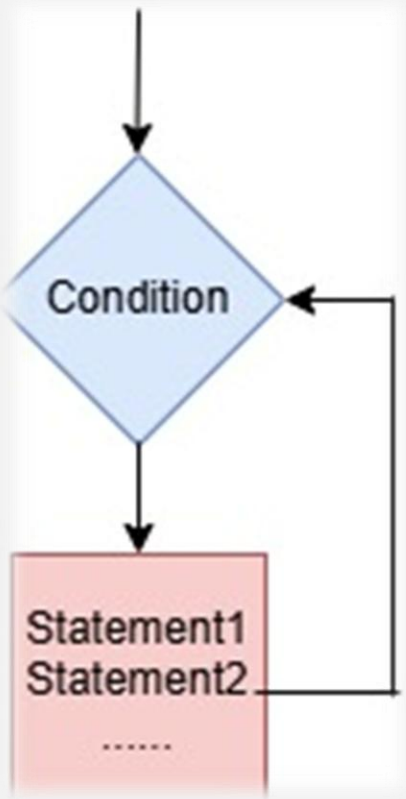
Looping Statements

- Loops allow repeated execution of a block of code.
- Useful for iteration and automation.
- Types
  - While loop
  - For loop

While loop

- Execute as long as condition is true
- Syntax

```
while condition:
    block of statement
else:
    else_block
```



Example

```
i=1
while i<=5:
    print(i,"MCA")
    i=i+1
```

i	Condition	Output
1<=5	True	1 MCA
2<=5	True	2 MCA
3<=5	True	3 MCA
4<=5	True	4 MCA
5<=5	True	5 MCA
6<=5	False (outside of the loop)	

```
1 MCA
2 MCA
3 MCA
4 MCA
5 MCA
```

What will be the output of the following code?

```
i=2
while i<10:
    print(i,end=' ')
    i=i+1
```

A. 2, 3, 4, 5, 6, 7, 8, 9, 10

B. 2, 3, 4, 5, 6, 7, 8, 9

C. 2, 3, 4, 5, 6, 7, 8, 9, 11

What will be the output of the following code?

```
i=2
while i<10:
    print(i,end=' ')
    i=i+1
```

A. 2, 3, 4, 5, 6, 7, 8, 9, 10

B. 2, 3, 4, 5, 6, 7, 8, 9

C. 2, 3, 4, 5, 6, 7, 8, 9, 11

for loop

- Used to iterate over a sequence.
- Block of code repeat for each item in a sequence object
- Syntax:

```
for item in sequence_object:
    body of the loop
else:
    else_block
```

Example

```
for i in "MCA":
    print(i, "-MCA")
```

Line 1 i	Line 2 $print$
M	M- MCA
C	C- MCA
A	A- MCA

```
M -MCA
C -MCA
A -MCA
```

$range$ function

- The $range()$ function is used to generate a sequence of numbers, commonly used with loops like for.
- $range(< start >, stop, < step >)$
- Start: starting value (default=0)
- Stop: ending value
- Step: increment (default=1)

```
for i in range(0,4):
    print(i,end=' ')

for i in range(4):
    print(i,end=' ')
```

```
for i in range(0,4, 2):
    print(i,end=' ')
```

What will be the output of the following code?

```
for i in range(2,8,2):
    print(i,end=' ')
```

A. 2, 4, 6

B. 2, 4, 6, 8

C. 0, 2, 4, 6

What will be the output of the following code?

```
for i in range(2,8,2):
    print(i,end=' ')
```

A. 2, 4, 6

B. 2, 4, 6, 8

C. 0, 2, 4, 6

for loop vs while loop

- For loop
  - Known number of iteration
  - Known as iterator

- While loop
  - Unknown iteration
  - Based on condition

Loop control statement

- Control the flow of loop

  - Break
  - Continue
  - pass

break Statement

- immediately exit the current for or while loop, transferring program control to the next statement after the loop.

```
name="123456"
for item in name:
    print(item)
    if item=='4':
        break
```

continue statement

- Control continues with the loop when control encounters *continue* statement.

- *pass* statement: do nothing

```
i=0
while i<=6:
    i=i+1
    if i%2==0:
        continue
    print(i,end=' |')
```

Nested loop

- A nested loop is a loop inside another loop.
  - Outer loop runs first
  - For each iteration of outer loop, inner loop runs completely

```
for i in range(3):
    print(i)
    for j in range(2):
        print(j,end=' ')
    print()
```

```
0
0 1
1
0 1
2
0 1
```

Nested loop

```
for i in range(3):
    for j in range(2):
        print(i,j,sep=' ')
```

Outer (i)	Inner (j)	Output
0	0	0 0
0	1	0 1
1	0	1 0
1	1	1 1
2	0	2 0
2	1	2 1

Quiz

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